

Experiment-7:

Write a program to implement Logistic Regression (LR) algorithm in python

Code:

```
# Logistic Regression
from google.colab import files
uploaded = files.upload()
import pandas as pd
import io
import zipfile
import os

# Get uploaded ZIP file
zip_file = list(uploaded.keys())[0]

# Extract ZIP
with zipfile.ZipFile(zip_file, 'r') as zip_ref:
    zip_ref.extractall("dataset")

# Find CSV file
for root, dirs, files in os.walk("dataset"):
    for file in files:
        if file.endswith(".csv"):
            csv_path = os.path.join(root, file)

# Read CSV
df = pd.read_csv(csv_path)
print("First 5 Rows:")
print(df.head())

# Features and Target
X = df.drop("target", axis=1)
y = df["target"]

# Train-Test Split
from sklearn.model_selection import train_test_split
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42
)
# Feature Scaling
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
# Logistic Regression
from sklearn.linear_model import LogisticRegression
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
# Prediction
y_pred = model.predict(X_test)
# Accuracy
from sklearn.metrics import accuracy_score
print("\nAccuracy:")
print(round(accuracy_score(y_test, y_pred) * 100, 2), "%")
# Confusion Matrix
from sklearn.metrics import confusion_matrix
cm = confusion_matrix(y_test, y_pred)
print("\nConfusion Matrix:")
print(cm)
# Classification Report
from sklearn.metrics import classification_report
print("\nClassification Report:")
print(classification_report(y_test, y_pred))
```

Output:

...

Choose Files

archive (7).zip

archive (7).zip(application/x-zip-compressed) - 3504 bytes, last modified: 8/4/2026 - 100% done

Saving archive (7).zip to archive (7) (1).zip

First 5 Rows:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	\
0	63	1	3	145	233	1	0	150	0	2.3	0	
1	37	1	2	130	250	0	1	187	0	3.5	0	
2	41	0	1	130	204	0	0	172	0	1.4	2	
3	56	1	1	120	236	0	1	178	0	0.8	2	
4	57	0	0	120	354	0	1	163	1	0.6	2	

	ca	thal	target
0	0	1	1
1	0	2	1
2	0	2	1
3	0	2	1
4	0	2	1

Accuracy:
85.25 %

Confusion Matrix:
[[25 4]
[5 27]]

Confusion Matrix:
[[25 4]
[5 27]]

Classification Report:				
	precision	recall	f1-score	support
0	0.83	0.86	0.85	29
1	0.87	0.84	0.86	32
accuracy			0.85	61
macro avg	0.85	0.85	0.85	61
weighted avg	0.85	0.85	0.85	61